

Software Requirements Specification (SRS)

Dropshipping Inventory & Order Orchestrator (Problem Statement #38)

1. Introduction

1.1 Purpose: This document specifies the functional and non-functional requirements for an automated e-commerce backend that ingests storefront webhooks, verifies supplier stock, splits multi-vendor orders, and tracks shipments.

1.2 Scope: The system serves two primary stakeholders: E-Store Owners (who manage orders and monitor fulfillment) and Supplier Partners (who receive purchase orders and report stock and tracking status).

1.3 Definitions: PO = Purchase Order; SKU = Stock Keeping Unit; Webhook = HTTP callback triggered by storefront events.

2. Overall Description

2.1 Product Perspective: A backend integration layer sitting between e-commerce storefront platforms (Shopify/WooCommerce) and multiple third-party dropshipping suppliers.

2.2 User Classes: E-Store Owner (admin-level access), Supplier Partner (restricted access to their own POs and tracking submissions).

2.3 Operating Environment: Cloud-hosted backend, REST API integrations with storefront and supplier platforms.

3. Functional & Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse inbound Shopify/WooCommerce order webhooks, verify supplier API inventory, and auto-dispatch supplier purchase orders.	High	Pass: Supplier PO triggered with customer delivery address; Fail: Order placed for out-of-stock supplier SKU.	Core order intake must never proceed on stale stock data.
FR-002	Functional	The system shall split a single customer order into multiple supplier sub-orders when line items originate from different suppliers.	High	Pass: A 3-item order from 2 suppliers generates exactly 2 supplier POs.	Multi-vendor orders can't be fulfilled by one PO.
FR-003	Functional	The system shall allow the E-Store Owner to view the real-time status of any order and its sub-orders.	Medium	Pass: Status updates reflect the latest supplier webhook within 5 seconds.	Owner needs visibility for customer service.
FR-004	Functional	The system shall allow a Supplier Partner to submit a tracking number against a specific supplier PO.	High	Pass: Tracking number is linked to the correct sub-order and visible to the store owner.	Enables order tracking and customer communication.
FR-005	Functional	The system shall automatically cancel and refund a sub-order if the assigned supplier does not confirm stock within a configurable timeout window.	Medium	Pass: Sub-order marked 'Cancelled, Refunded' after timeout.	Prevents orders from being stuck with unresponsive suppliers.

NFR-001	Nonfunctional	Inbound webhook processing must acknowledge receipt with HTTP 200 within 200 ms.	High	Pass: Benchmarking confirms target latency under peak load.	Storefront platforms retry or disable webhooks on slow acknowledgements.
NFR-002	Nonfunctional	The system shall encrypt all supplier API credentials at rest and in transit using AES-256 and TLS 1.2+.	High	Pass: Security scan confirms no plaintext credentials in storage or logs.	Protects sensitive third-party integration secrets.

4. External Interface Requirements

- Storefront webhook API (Shopify/WooCommerce)
- Supplier inventory/PO APIs
- Payment gateway API

5. Assumptions and Dependencies

- Suppliers expose a REST API for stock verification and PO acceptance
- Storefront platforms support standard webhook retry policies